The Likelihood of a Computed Dynamic Model: Proposition 2 of Fernández-Villaverde, Rubio-Ramírez and Santos, Corrected

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Abstract

Proposition 2 of Fernández-Villaverde et al. [2006] bounds the gap between the exact and the computed likelihood of a dynamic model by a term linear in the sample length, $TB\chi\delta$, with constants that do not depend on T. Ackerberg et al. [2009], following Geweke [2007], show by a Gaussian counterexample that this is false. We locate the error in a single step of the proof: a difference of two T-fold products of densities is bounded as if the other $T-1$ factors were absent, so that the "constant" B is really M^{T-1}, where M bounds the density. We then show that the proposition survives, with the same shape, once it is stated for the *logarithm* of the likelihood: under the paper's lower bound on densities, a pathwise version of its contraction condition, and a Lipschitz condition on the log density in the initial state, $|\log L_j(y^T) - \log L(y^T)| \le (T\chi/\xi + K/(1-\alpha))\,\delta$. The dependence on T is sharp. The average log-likelihood error is therefore $O(\delta)$ uniformly in T, which is the quantity Ackerberg, Geweke and Hahn identify as relevant for estimation. The core of the argument is formalised in Lean 4 with Mathlib.

1 Setting

Fernández-Villaverde et al. [2006] (FRS) study a state-space model

$$S_t = \varphi(S_{t-1}, W_t; \gamma), \qquad Y_t = g(S_t, V_t; \gamma),$$

with compact supports, a unique invariant distribution μ^* of the state, and computed approximations φ_j, g_j satisfying $\|\varphi_j - \varphi\| \le \delta$ and $\|g_j - g\| \le \delta$ in the sup norm. The shock W_t is split as (W_{1t}, W_{2t}), where (W_{2t}, V_t) is recovered from the data by inverting the system.

*The Lean development is `LeanEconomics/Econometrics/ApproxLikelihood.lean` in `https://github.com/LeanEconomics/LeanEconomics`. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result marked with a formal statement was checked by the Lean kernel and uses only Lean's three standard axioms.

Write $\omega = W_1^T$ for the unobserved shock path and

$$p_t(s,\omega) = p(y_t \mid W_1^t = \omega_{\le t},\, S_0 = s,\, y^{t-1};\gamma)$$

for the conditional density of the t-th observation, with q_t its counterpart under (φ_j, g_j). Then

$$L(y^T) = \iint \prod_{t=1}^{T} p_t(s,\omega)\, Q(\mathrm{d}\omega)\, \mu^*(\mathrm{d}s), \qquad L_j(y^T) = \iint \prod_{t=1}^{T} q_t(s,\omega)\, Q(\mathrm{d}\omega)\, \mu_j^*(\mathrm{d}s),$$

where Q is the law of W_1^T and μ_j^* the invariant distribution of the computed chain. FRS's Assumptions 4 and 8 give a lower bound $p_t, q_t \ge \xi > 0$; their Lemma 6 gives $|q_t - p_t| \le \chi\delta$; and their Condition 1 is the mean contraction $\int \|\varphi(S,W) - \varphi(S',W)\|\, \mathrm{d}Q(W) \le \alpha\|S - S'\|$ with $\alpha < 1$.

FRS's Proposition 2. Under Assumptions 1–11 and Condition 1, there are constants B and L such that, for all T,

$$|L_j(y^T) - L(y^T)| < \Big(TB\chi + \frac{L}{1-\alpha}\Big)\delta.$$

2 Where the proof fails

FRS split the gap as $|L_j - L| \le |\int F\,\mathrm{d}\mu_j^* - \int F\,\mathrm{d}\mu^*| + |\int (F_j - F)\,\mathrm{d}\mu_j^*|$, with $F = \int \prod_t p_t\,\mathrm{d}Q$ and $F_j = \int \prod_t q_t\,\mathrm{d}Q$. Their bound (14) on the second term is $TB\chi\delta$. But the telescoping identity

$$\prod_{t=1}^{T} q_t - \prod_{t=1}^{T} p_t = \sum_{t=1}^{T} \Big(\prod_{s<t} q_s\Big)(q_t - p_t)\Big(\prod_{s>t} p_s\Big)$$

places $T-1$ density factors beside each error $q_t - p_t$. With M a bound on the densities the honest bound is $TM^{T-1}\chi\delta$, and M^{T-1} is not a constant: it grows exponentially when $M > 1$ and vanishes exponentially when $M < 1$. This is exactly the factor $(\sigma\sqrt{2\pi})^{-(T-1)}$ in Ackerberg, Geweke and Hahn's sharp bound for the Gaussian location model. The step cannot be repaired with any T-free constant:

Proposition 1. *Let $M > 1$ and $\varepsilon > 0$. There is no B such that $|(M+\varepsilon)^T - M^T| \le TB\varepsilon$ for every $T \in \mathbb{N}$.*

Formal statement: `not_exists_linear_bound_prod_sub`.

Proof. By Bernoulli's inequality $(M+\varepsilon)^{n+1} - M^{n+1} \ge (n+1)\varepsilon M^n$, so the bound would force $M^n \le B$ for all n. □

Two further steps let constants depend on T without comment. The Lipschitz constant of F in S_0, used in (12), is "different for each t, but since t is finite, we can set a global L"; the proposition is then stated "for all T". And Lemma 6's χ is asserted for all t "since the

model is stationary", although the recovered (v_t, s_t, w_{2t}) depend on a t-fold composition of the transition. We return to both in Section 4.

3 The corrected proposition

The accumulation of error that FRS describe is real, but it is *multiplicative*: it holds for the log likelihood, or equivalently the likelihood ratio, and not for the level.

Hypothesis 1 (Uniform per-period error)**.** $\xi \le p_t(s,\omega)$, $\xi \le q_t(s,\omega)$ and $|q_t(s,\omega) - p_t(s,\omega)| \le \varepsilon$ for all t, s, ω, with $\xi > 0$. In FRS's notation $\varepsilon = \chi\delta$.

Hypothesis 2 (Log-Lipschitz in the initial state)**.** There are $\ell_t \ge 0$ with $p_t(s',\omega) \le e^{\ell_t d(s,s')} p_t(s,\omega)$ and likewise for q_t, for all s, s', ω. Write $K = \sum_{t=1}^{T} \ell_t$.

Hypothesis 3 (Coupled initial states)**.** There is a probability measure P on a space X and maps $a, b : X \to S$ whose laws are μ^* and μ_j^*, with $d(a(x), b(x)) \le \rho$ for P-almost every x.

Proposition 2 (Proposition 2′)**.** *Under Hypotheses 1–3, and integrability of the likelihood integrands,*

$$e^{-D} L(y^T) \le L_j(y^T) \le e^{D} L(y^T), \qquad D = \frac{T\varepsilon}{\xi} + K\rho,$$

equivalently $|\log L_j(y^T) - \log L(y^T)| \le D$.

Formal statement: `likelihood_le_exp_mul, abs_log_likelihood_sub_le`.

Proof. Step 1 (one factor). If $\xi \le p$ and $|q - p| \le \varepsilon$ then $q \le p + \varepsilon \le p(1 + \varepsilon/\xi) \le e^{\varepsilon/\xi} p$.
Formal statement: `le_exp_div_mul`.

Step 2 (the product; this replaces FRS's (14)). Multiplying Step 1 over t, $\prod_t q_t \le e^{T\varepsilon/\xi} \prod_t p_t$ pointwise in (s, ω).
Formal statement: `prod_le_exp_card_mul_prod`.

Step 3 (moving the initial state). Multiplying Hypothesis 2 over t, $\prod_t p_t(s',\omega) \le e^{Kd(s,s')} \prod_t p_t(s,\omega)$.

Step 4 (integrating; this replaces FRS's (12)). Combining Steps 2 and 3 and integrating over ω, $\int \prod_t q_t(s',\cdot)\,\mathrm{d}Q \le e^{T\varepsilon/\xi + Kd(s,s')} \int \prod_t p_t(s,\cdot)\,\mathrm{d}Q$ for all s, s'. Put $s = a(x)$, $s' = b(x)$, use $d(a(x), b(x)) \le \rho$ almost surely, and integrate against P: $L_j \le e^D L$.
Formal statement: `lik_le_exp_mul_lik`.

The reverse inequality is the same argument with the roles of the two models exchanged. Since the densities are at least ξ and Q, P are probability measures, $L, L_j \ge \xi^T > 0$, and taking logs gives the second form. □

Remark 3 (Sharpness in T). With every exact density equal to M and every computed one to $M + \varepsilon$, $\log L_j - \log L = T \log(1 + \varepsilon/M)$ exactly. The bound cannot grow more slowly than linearly in T. In Ackerberg, Geweke and Hahn's location model restricted to a compact support, taking every y_t at the same point $\gamma + c$ gives $\log L_j - \log L = T(c\delta - \delta^2/2)/\sigma^2$, again linear in T.

Formal statement: `log_prod_div_prod_const`.

4 The hypotheses

Hypothesis 1 is FRS's Assumptions 4 and 8 and the conclusion of their Lemma 6, stated uniformly in t. Hypotheses 2 and 3 replace their Condition 1 and the Santos–Peralta-Alva bound used in (12). We now say when they hold.

The coupling. Strengthen Condition 1 from contraction in mean to *pathwise* contraction: $d(\varphi(s,w),\varphi(s',w)) \le \alpha\, d(s,s')$ for every shock w. Drive the exact and computed chains by the same shocks. One step gives

$$d(\varphi_j(s',w),\varphi(s,w)) \le \alpha\, d(s',s) + \delta,$$

so the band $D_\rho = \{(s,s') : d(s,s') \le \rho\}$, $\rho = \delta/(1-\alpha)$, is invariant under the joint chain, and from any starting pair $d(s_{j,t},s_t) \le \alpha^t d(s_{j,0},s_0) + \delta/(1-\alpha)$.
Formal statement: `dist_step_le, dist_step_le_of_le, dist_iterate_le`.
On a compact state space with continuous transitions the joint chain restricted to the compact band D_ρ has an invariant probability measure (Krylov–Bogolyubov). Its marginals are invariant for each chain, so by uniqueness they are μ^* and μ_j^*, and that measure is the P of Hypothesis 3. This last step is not formalised: Mathlib does not yet have Krylov–Bogolyubov. It also needs uniqueness of the computed invariant distribution, which FRS do not assume.

The Lipschitz constants, and Lemma 6 uniformly in t. Consider the case $\dim V_t = \dim Y_t$, so that W_2 is empty and V_t alone is recovered from the data: s_t is then a function of (s_0,ω) alone and $v_t = g^{-1}(y_t;s_t)$. Pathwise contraction gives $d(s_t(s_0,\omega),s_t(s_0',\omega)) \le \alpha^t d(s_0,s_0')$
Formal statement: `dist_iterate_exact_le`.
so if $\log p_t$ is c-Lipschitz in the current state s_t (true when the density of V and the inverse of g are Lipschitz and the density is bounded below), then $\ell_t \le c\alpha^t$ and $K \le c/(1-\alpha)$, uniformly in T. In the same case $d(s_{j,t},s_t) \le \delta/(1-\alpha)$ from a common initial state, which gives Hypothesis 1 with $\varepsilon = \chi\delta$ for a χ that does not depend on t. These two sentences are an argument on paper; only the pathwise distance bounds are formal.

When W_2 is not empty, the recovered shocks depend on the data and on the whole past path, and Hypotheses 1 and 2 become a stability condition on the inverted filter. It is not implied by FRS's Condition 1, and we have not found a primitive condition that delivers it.

Corollary 4 (Constants uniform in T)**.** *If in addition $\ell_t \le c\alpha^t$ and $\rho = \delta/(1-\alpha)$, then for every T*

$$|\log L_j(y^T) - \log L(y^T)| \le \Big(\frac{T\chi}{\xi} + \frac{c}{(1-\alpha)^2}\Big)\delta,$$

and so

$$\frac{1}{T}|\log L_j(y^T) - \log L(y^T)| \le \Big(\frac{\chi}{\xi} + \frac{c}{T(1-\alpha)^2}\Big)\delta.$$

Formal statement: `abs_log_likelihood_sub_le_of_geometric`.

This is FRS's bound, $T\cdot\text{const}\cdot\delta + \text{const}\cdot\delta/(1-\alpha)$, with the level replaced by the log.

5 What survives of FRS's conclusions

- *Error accumulates with T.* True of the log likelihood, at exactly the linear rate FRS describe. In relative terms $|L_j/L - 1| \le e^D - 1 \approx D$ while D is small, so FRS's message "the error is of order $T\delta$" is correct for the *relative* error of the likelihood. Their statement for the absolute error is false.
- *Likelihood ratios and Metropolis–Hastings (FRS Corollary 1).* For two parameter values, $L_j(\gamma')/L_j(\gamma)$ lies within a factor $e^{\pm 2D}$ of $L(\gamma')/L(\gamma)$, which bounds the error in every acceptance probability.
- *Posteriors.* If the hypotheses hold uniformly in γ, the marginal likelihoods are within $e^{\pm D}$ of each other, so the posterior density ratio $p_j(\gamma \mid y^T)/p(\gamma \mid y^T)$ lies in $[e^{-2D}, e^{2D}]$, and the total variation distance between the two posteriors is at most $e^{2D} - 1$. This is not formalised.
- *Estimation.* The average log likelihood $\frac{1}{T}\log L_j$ is within $O(\delta)$ of $\frac{1}{T}\log L$, uniformly in T (Corollary 4). This is the object Ackerberg et al. [2009] identify as relevant for the maximum likelihood estimator, and it agrees with their conclusion that approximation error of order δ moves the estimator's probability limit by $O(\delta)$, not by more as T grows. A sup bound on the objective alone gives only $O(\sqrt{\delta})$ for the maximiser under quadratic identification; their $O(\delta)$ rate uses closeness of derivatives as well.

What does not survive is FRS's conclusion that linearisation "is due to fail as the sample size grows" because the likelihood levels diverge. The levels of both likelihoods go to zero or infinity like M^T anyway. What grows is the log-likelihood gap, and it grows no faster than the log likelihood itself, so the average stays $O(\delta)$.

6 The formal development

Result	Lean name
One factor, additive to multiplicative error	`le_exp_div_mul`
Corrected product step (14)	`prod_le_exp_card_mul_prod`
FRS's step (14) cannot hold (Prop. 1)	`not_exists_linear_bound_prod_sub`
Sharpness in T	`log_prod_div_prod_const`
Coupled chain: one step, invariant band	`dist_step_le`, `dist_step_le_of_le`
Coupled chain: path bound	`dist_iterate_le`
Exact chain forgets its initial state	`dist_iterate_exact_le`
Likelihood bounded below by ξ^T	`pow_le_lik`
Proposition 2′, one direction	`lik_le_exp_mul_lik`
Proposition 2′, two-sided form	`likelihood_le_exp_mul`
Proposition 2′, log form	`abs_log_likelihood_sub_le`
Corollary 4	`abs_log_likelihood_sub_le_of_geometric`

The likelihood is formalised as $\int_X \int_\Omega \prod_t p_t(a(x), \omega)\, \mathrm{d}Q\, \mathrm{d}P$ for an arbitrary finite index set of periods, a pseudometric state space and arbitrary measure spaces of shock paths and

couplings. Using the coupling space X rather than the marginal μ^* avoids a measurability hypothesis on $s \mapsto \int \prod_t p_t(s, \cdot)\,\mathrm{d}Q$; when $a = \mathrm{Prod.fst}$ and that map is measurable the two agree by change of variables. Not formalised: the existence of the coupling (Krylov–Bogolyubov), the verification of Hypotheses 1 and 2 from primitives in the case with no W_2, and the posterior bounds.

References

Daniel Ackerberg, John Geweke, and Jinyong Hahn. Comments on "Convergence properties of the likelihood of computed dynamic models". *Econometrica*, 77(6):2009–2017, 2009. doi: 10.3982/ECTA7669.

Jesús Fernández-Villaverde, Juan F. Rubio-Ramírez, and Manuel S. Santos. Convergence properties of the likelihood of computed dynamic models. *Econometrica*, 74(1):93–119, 2006.

John Geweke. Convergence properties of the likelihood of computed dynamic models: Comment. Unpublished manuscript, University of Iowa, 2007.